MIL-R-39009/2A
22 April 1970
SUPERSEDING
MIL-R-39009/2
30 August 1967

MILITARY SPECIFICATION

# RESISTORS, FIXED, WIRE-WOUND (POWER TYPE, CHASSIS MOUNTED), ESTABLISHED RELIABILITY, STYLES RER40, RER45, RER50, AND RER55

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 This specification covers the detail requirements for styles RER40, RER45, RER50 and RER55 resistors. These resistors shall be noninductively wound by one of the methods specified in 3.2.2.

## 2. APPLICABLE DOCUMENTS

2.1 The following document, of the issue in effect on date of invitation for bids or request for proposal, forms a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-R-39009 - Resistors, Fixed, Wire-wound (Power Type, Chassis Mounted), Established Reliability, General Specification for.

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

## 3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with MIL-R-39009, and as specified herein.

3.2 Design and construction. Resistors shall be of the design, construction, and physical dimensions specified on figure 1, and as specified herein.

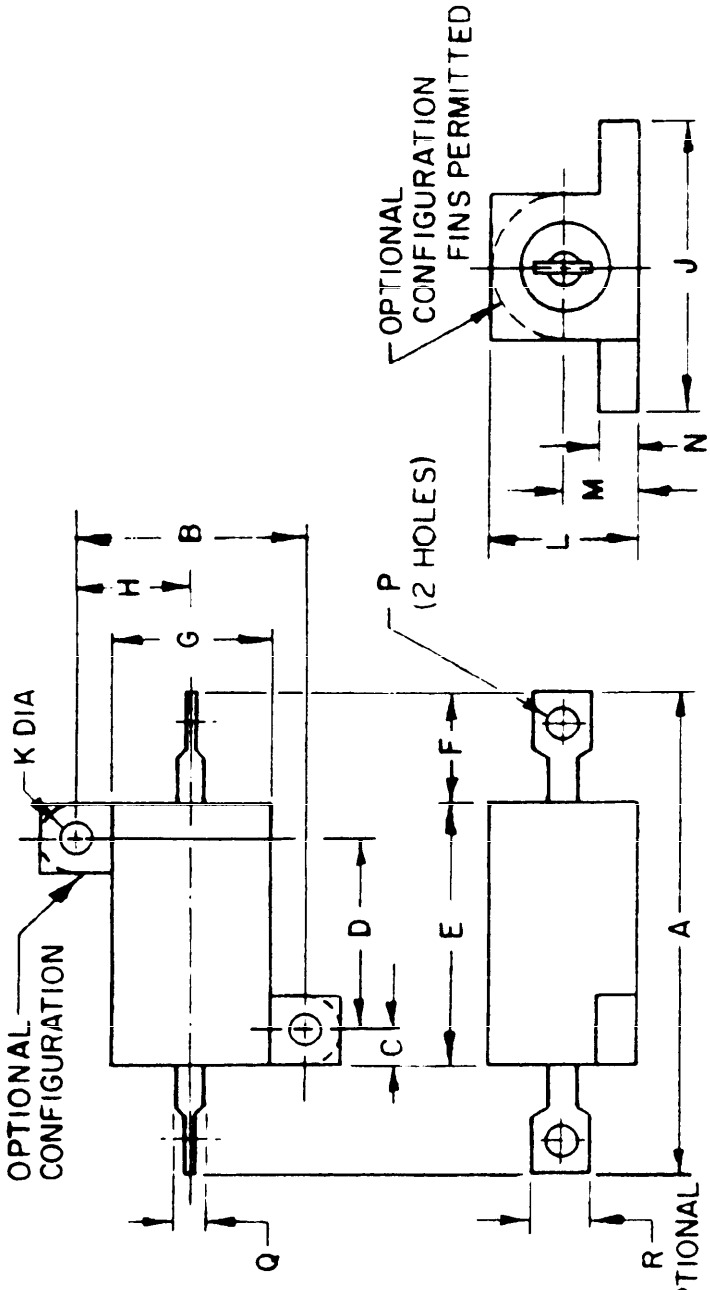
3.2.1 Weight. The maximum weight shall be as specified in table I.

TABLE I. Weight.

| Resistor style | Weight |
|---|---|
| | Grams, maximum |
| RER40 | 3.3 |
| RER45 | 8.8 |
| RER50 | 16.5 |
| RER55 | 35.0 |

FSC 5905

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| Resistor style | A ±.062 (1.57) | B ±.010 (.25) | C ±.031 (.79) | D ±.010 (.25) | E ±.062 (1.57) | F ±.062 (1.57) | G ±.062 (1.57) | H ±.031 (.79) | J ±.031 (.79) | K ±.005 (.13) | L ±.031 (.79) | M ±.062 (1.57) | N ±.031 (.79) | P ±.005 (.13) | Q min AWG | R min |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| RER40 | 1.125 (28.58) | .490 (12.45) | .078 (1.98) | .444 (11.28) | .600 (15.24) | .266 (6.76) | .334 (8.48) | .245 (6.22) | .646 (16.41) | .093 (2.36) | .320 (8.13) | .133 (3.38) | .065 (1.65) | .050 (1.27) | 16 | .085 (2.16) |
| RER45 | 1.375 (34.93) | .625 (15.88) | .094 (2.39) | .562 (14.27) | .750 (19.05) | .312 (7.92) | .438 (11.13) | .312 (7.92) | .812 (20.62) | .094 (2.39) | .406 (10.31) | .203 (5.16) | .094 (2.39) | .085 (2.16) | 12 | .140 (3.56) |
| RER50 | 1.938 (49.23) | .781 (19.84) | .172 (4.37) | .719 (18.26) | 1.062 (26.97) | .438 (11.13) | .531 (13.49) | .391 (9.93) | 1.094 (27.79) | .125 (3.18) | .562 (14.27) | .281 (7.14) | .094 (2.39) | .085 (2.16) | 12 | .140 (3.56) |
| RER55 | 2.781 (70.64) | .844 (21.44) | .188 (4.78) | 1.562 (39.67) | 1.938 (49.23) | .438 (11.13) | .594 (15.09) | .422 (10.72) | 1.156 (29.36) | .125 (3.18) | .625 (15.88) | .312 (7.92) | .094 (2.39) | .085 (2.16) | 12 | .140 (3.56) |

NOTES: 1. All dimensions in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Millimeters are in parentheses.

FIGURE 1. Styles RER40, RER45, RER50, and RER55 (non-inductively-wound) resistors.



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3.2.2 Winding. In order to minimize inductance, resistors shall be wound by one of the following methods:

(a) Ayrton-Perry.
(b) Bifilar.

* 3.3 Power rating. The power rating shall be as specified in table II, based on full-load operation at an ambient temperature of 25° C, when mounted on the chassis specified in 4.4 (see 6.4).

TABLE II. Power rating.

| Resistor style | Power rating |
|---|---|
| | Watts |
| RER40 | 5 |
| RER45 | 10 |
| * RER50 | 20 |
| RER55 | 30 |

3.3.1 Power rating (free air). The free air (resistor not mounted on a chassis) power rating based on full load operation at 25° C shall be as specified in table III.

TABLE III. Power rating (free air).

| Resistor style | Power rating |
|---|---|
| | Watts |
| RER40 | 3 |
| RER45 | 6 |
| RER50 | 8 |
| RER55 | 10 |

3.4 Resistance. The minimum and maximum nominal resistance values shall be as specified in table IV.

TABLE IV. Minimum and maximum nominal resistance values.

| Resistor style | Resistance value | |
|---|---|---|
| | Minimum | Maximum |
| | Ohm | Ohms 1/ |
| RER40 | 1 | 1,650 |
| RER45 | 1 | 2,800 |
| RER50 | 1 | 6,040 |
| RER55 | 1 | 19,600 |

1/ Based on use of 0.001-inch nominal diameter wire.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-R-39009, and as specified herein.



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4.2 Dielectric withstanding voltage.

4.2.1 At atmospheric pressure. The magnitude of the test voltage shall be as specified in table V.

TABLE V. Dielectric-withstanding-test voltages at atmospheric pressure.

| Resistor style | Dielectric-withstanding-test voltage |
|---|---|
| | Volts, (rms) |
| RER40 | 1,000 |
| RER45 | 1,000 |
| RER50 | 1,000 |
| RER55 | 2,000 |

4.2.2 At reduced barometric pressure. The magnitude of the test voltage shall be 500 volts.

4.3 Terminal strength. The direct pull shall be as specified in table VI.

TABLE VI. Direct pull.

| Resistor style | Direct pull |
|---|---|
| | Pounds |
| RER40 | $5\ ^{+0}_{-1/4}$ |
| RER45 | $5\ ^{+0}_{-1/4}$ |
| RER50 | $10\ ^{+0}_{-1/2}$ |
| RER55 | $10\ ^{+0}_{-1/2}$ |

4.4 Life. The chassis dimensions shall be as specified in table VII.

TABLE VII. Chassis dimensions.

| Resistor style | Length, width, and height | Thickness |
|---|---|---|
| | Inches | Inch |
| RER40 | 6 x 4 x 2 | 0.04 |
| RER45 | 6 x 4 x 2 | 0.04 |
| RER50 | 7 x 5 x 2 | 0.04 |
| RER55 | 7 x 5 x 2 | 0.04 |

5. PREPARATION FOR DELIVERY

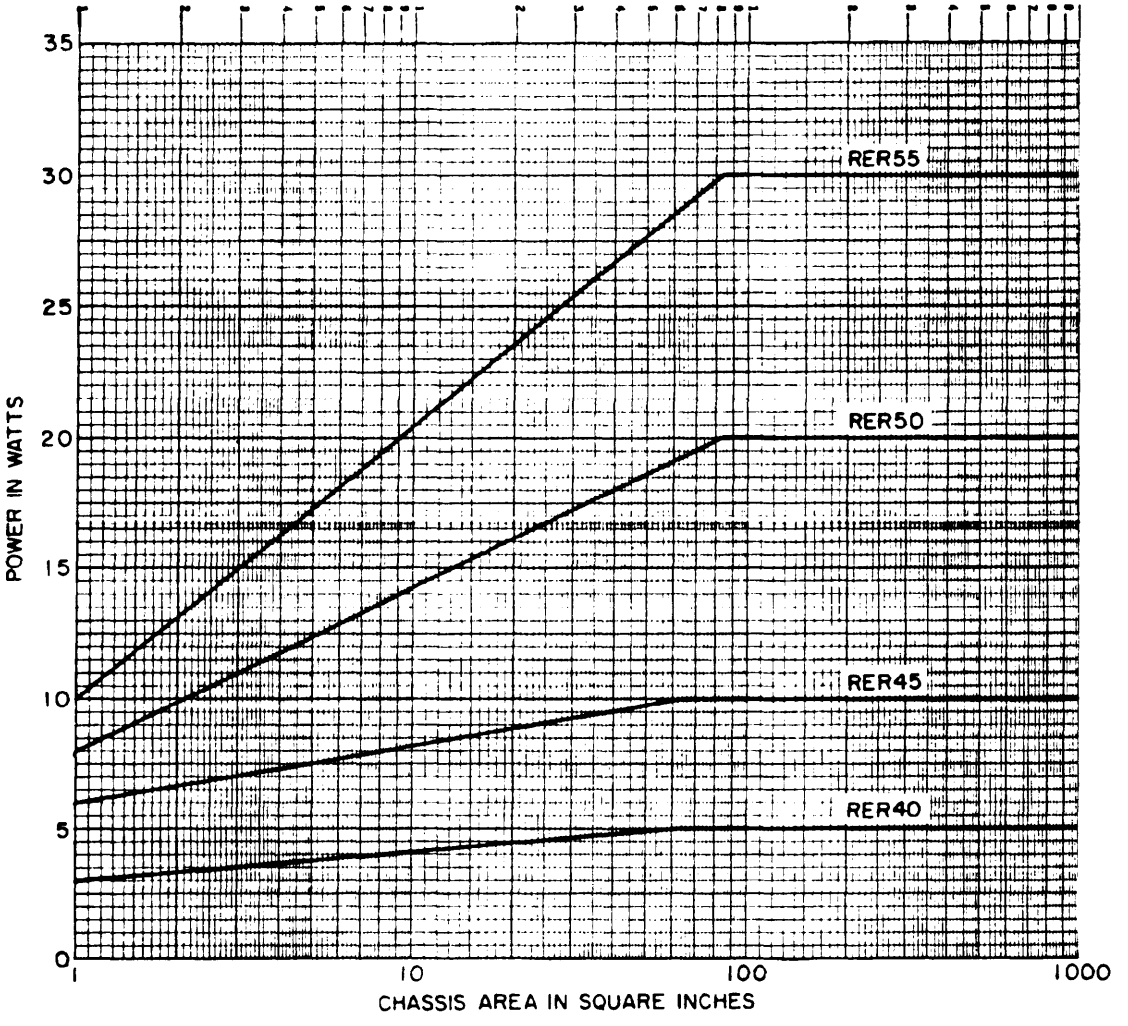
5.1 Preparation for delivery shall be in accordance with MIL-R-39009.

6. NOTES

6.1 Notes. In addition to the notes specified herein, the notes specified in MIL-R-39009 are applicable to this specification.



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NOTE:
The chassis derating curves are based on the full power ratings at an ambient temperature of 25°C. These curves are independent of the temperature derating curves.

FIGURE 2. Chassis-area derating curves.



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* 6.2 MIL-R-18546/1 substitution data. Resistors of this specification, regardless of failure rate, are substitutes for resistors of the same resistance value and characteristic N of the inactivated specification MIL-R-18546/1 as follows:

| MIL-R-18546/1 style characteristic N (non-inductively wound) | Substitute MIL-R-39009/2 style |
|---|---|
| RE75 | RER55 |
| RE70 | RER50 |
| RE65 | RER45 |
| RE60 | RER40 |

* 6.2.1 Interchangeability. The increase in power rating for RER50 by this revision does not require the assignment of new Federal Stock Numbers for items previously described as RER50, 15 watt rating. Items in this specification are mutually interchangeable with items of the same type designation procured under MIL-R-39009/2A.

6.3 Chassis-area derating curves. Figure 2 may be used for new design information.

* 6.4 Change in power rating RER50. The revision of style RER50 to 20 watts from the previous 15-watt rating is based on qualification of similar style RE70 (characteristic N) under MIL-R-18546. In addition, the accumulated failure rate component hours at 15 watts have been halved for qualification failure rate level retention. Until sufficient hours are established at 20 watts, the Qualified Products List will continue to list FR levels at the previous 15-watt rating, and, in addition, the FR level at the 20-watt rating.

* 6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL, ME, AT, AV
Navy - EC, AS, OS
Air Force - 11, 17, 85
DSA - ES

User activities:
Army - MI, MU
Navy - MC, CG
Air Force - 19

Preparing activity:
Army - EL

Agent:
DSA - ES

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